

CANONICAL LESSON

Beds, Staff, Supplies: Forecasting Service Capacity

Decision-centred foundations for accountable sector intelligence

Production Unit	PU-B07-C22
Notebook	V7_B_N02
Status	Controlled educational draft v0.1
Data boundary	Synthetic instructional data only

Central decision: Forecast beds, staff, medicines, and service demand with time-aware baselines, uncertainty, constraints, and operational ownership.

Learning outcomes

Why this matters

Nominal bed counts can conceal unavailable staff, supplies, equipment, or referral capacity.

The analytical workflow is useful only when evidence, uncertainty, authority, and action are connected explicitly. The model supports professional judgement; it does not replace accountable institutional authority.

Worked cross-sector example

A facility forecast links expected admissions to staffed beds, shift coverage, oxygen, and supply availability.

Boundary: The example supports learning and review. It does not constitute an official statistic, eligibility determination, enforcement action, clinical judgement, or operational authorization.

Practice and reflection

- 1.
- Use rolling-origin evaluation.
- 2.
- Add separate emergency and elective demand.

- 3.
- Model staff-hours as a constraint.
- 4.
- Explain why maximizing occupancy can be unsafe.

Answer key

- 1.
- Reforecast from successive dates using only available history and score each horizon.
- 2.
- Use distinct concepts, arrival processes, and intervention rules; do not aggregate incompatible demand.
- 3.
- Convert workload through documented skill-mix and hours assumptions, then optimize under labour rules.
- 4.
- High occupancy removes surge buffer, increases delay and infection/quality risks, and ignores case mix and staffing.

Operational readiness checklist

- Purpose and decision owner are named.
- Evidence definitions, provenance, and lawful access are documented.
- Uncertainty and abstention are visible.
- Human review, escalation, monitoring, incident response, and retirement are assigned.
- Equity, accessibility, security, and sector-specific approval are complete.